

Two-Sample Tests Use the information provided in Exercises 7–8. State the null and alternative hypotheses for testing for a significant difference in means, calculate the pooled estimate of σ^2 , the associated degrees of freedom, and the observed value of the t statistic. What is the rejection region using $\alpha = .05$? What is the p -value for the test? What can you conclude from these data?

7. Population 1	12	3	8	5	
Population 2	14	7	7	9	6

8. Population	1	2
Sample Size	16	13
Sample Mean	34.6	32.2
Sample Variance	4.8	5.9

$$\begin{aligned}
 7. \quad n_1 &= 4, \bar{X}_1 = 7, s_1^2 = 15.333 \\
 n_2 &= 5, \bar{X}_2 = 8.6, s_2^2 = 10.3 \\
 s^2 &= \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \approx 12.457 \\
 df &= 4 + 5 - 2 = 7
 \end{aligned}$$

$$H_0 = \mu_1 - \mu_2 = 0, H_a = \mu_1 - \mu_2 \neq 0$$

$$8. \quad s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \approx 5.289$$

$$df = 16 + 13 - 2 = 27$$

$$t^* = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s^2(\frac{1}{n_1} + \frac{1}{n_2})}} = \frac{34.6 - 32.2}{\sqrt{5.289(\frac{1}{16} + \frac{1}{13})}} \approx 2.795$$

$$t_{27, 0.025} = 2.052, \quad 2.795 > 2.052 \Rightarrow \text{reject } H_0$$

$$t > t_{27, 0.025} \Rightarrow 2P < 0.005 \Rightarrow P\text{-value} < 0.01 < 0.05$$

$$H_0: \mu_1 - \mu_2 = 0, H_a: \mu_1 - \mu_2 \neq 0$$

$$t^* = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2(\frac{1}{n_1} + \frac{1}{n_2})}} \approx -0.675$$

$$t_{7, 0.025} = 2.365$$

$$-0.675 > -2.365 \Rightarrow \text{Not reject}$$

$$t_{7, 0.1} > t$$

$$\Rightarrow 0.1 < P$$

$$\Rightarrow 0.2 < 2P$$